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AI Skills Lab – COE National Hackathon

Theme: AI Beyond Chatbots — Building Intelligent Systems That Reason, Act & Impact

Problem Statement 1 — Autonomous AI for Disaster Response

Track: Agentic AI + Multimodal AI + Social Impact

Background

During floods, earthquakes, landslides and other disasters, authorities receive information from multiple sources such as emergency calls, social media, satellite imagery, weather reports, IoT sensors and field personnel. The information is often fragmented, noisy and arrives at different times.

Human teams must quickly determine **what is happening, where help is required, which resources should be deployed and what actions should be prioritized** .

Challenge

Develop an **AI-powered autonomous disaster-response system** capable of collecting information from multiple sources, reasoning over the situation and recommending or executing appropriate response actions.

The system should be able to:

- Process **text, images, audio, maps or sensor data** .
- Identify and prioritize emergency incidents.
- Determine the severity and geographical location of incidents.
- Recommend allocation of emergency resources.
- Generate situation reports for authorities.
- Continuously update recommendations as new information arrives.
- Provide an explainable justification for critical decisions.

Expected Outcome

A working prototype demonstrating an **AI agent or multi-agent system** that can move from:

Observe → Understand → Reason → Prioritize → Act

Teams may use simulated emergency data if real-time data is unavailable.

Innovation Opportunities

- Multi-agent AI
 - Computer vision
 - LLMs
 - GIS
 - Satellite imagery
 - RAG
 - IoT
 - Predictive analytics
 - Human-in-the-loop decision making
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Problem Statement 2 – AI Guardian: Detecting and Defending Against AI-Generated Threats

Track: AI Security + Responsible AI

Background

Generative AI has significantly lowered the barrier to creating convincing fake content. Deepfake videos, synthetic voices, manipulated images, phishing messages and AI-generated misinformation can be used for fraud, impersonation and social engineering.

At the same time, AI applications themselves are vulnerable to attacks such as **prompt injection, data leakage, model manipulation and malicious inputs** .

Challenge

Build an **AI security system capable of identifying, analyzing and responding to AI-generated or AI-assisted threats** .

The system may focus on one or more of the following:

- Deepfake image/video detection

- Synthetic voice detection
- AI-generated phishing detection
- Malicious prompt detection
- Prompt-injection defense
- AI-generated misinformation detection
- Identity impersonation detection
- Sensitive-data leakage prevention

The system should not merely classify an input as “*safe*” or “*unsafe*” . It should provide:

Detection → Risk Assessment → Explanation → Recommended Action

Expected Outcome

Teams should develop a prototype that can:

1. Accept potentially malicious or manipulated content.
2. Analyze the content using AI/ML techniques.
3. Assign a risk/confidence score.
4. Explain the major indicators responsible for the decision.
5. Recommend an appropriate mitigation action.

Innovation Opportunities

- Multimodal AI
- Computer vision
- NLP
- Audio analysis
- LLM security
- Adversarial ML
- Explainable AI
- Cybersecurity
- Digital forensics

Special Challenge

Teams may optionally demonstrate how their system performs against **previously unseen attack/manipulation techniques** .

Problem Statement 3 — AI for Bharat: Intelligent Multilingual Public-Service Assistant

Track: AI for Bharat + Multimodal AI + Inclusive Technology

Background

India's linguistic diversity creates a major challenge for digital public services. Many AI systems work extremely well in English but provide limited support for regional languages, code-mixed communication, speech-based interaction and users with limited digital literacy.

A citizen may communicate using:

Hindi + English + regional language + voice + images

rather than formal written English.

Challenge

Develop a **multilingual and multimodal AI system designed for Indian users** that can understand user intent and help them access information or services.

The solution should support multiple modalities such as:

Voice → Text → Reasoning → Response → Voice

and/or

Image → Understanding → Reasoning → Action

Potential applications include:

- Government service assistance
- Agricultural advisory
- Student support
- Healthcare information
- Legal-awareness assistance
- Women's safety
- Financial-literacy assistance
- Rural entrepreneurship
- Accessibility for elderly or differently abled users

Expected Outcome

The system should demonstrate:

- Support for **at least two Indian languages** .
- Handling of code-mixed queries.
- Voice-based interaction where applicable.
- Retrieval of information from trusted sources.
- Context-aware responses.
- Appropriate handling of uncertain or unavailable information.
- Clear indication of the source/reasoning behind important answers.

Innovation Opportunities

- Indic language models
- Speech recognition
- Text-to-speech
- RAG
- Multimodal LLMs
- OCR
- Translation
- Knowledge graphs
- Accessibility technologies

Special Challenge

The system should demonstrate performance on **realistic Indian-language queries rather than only carefully written benchmark questions** .

Problem Statement 4 — Autonomous AI for Sustainable Smart Campus

Track: AI + Sustainability + IoT + Agentic AI

Background

Educational campuses, offices and industrial facilities consume significant amounts of electricity and water. Much of this consumption is difficult to optimize because information is distributed across different systems.

For example:

- Electricity consumption varies by building.
- Classrooms may remain powered when unoccupied.
- Water leakage may go unnoticed.
- HVAC systems may operate inefficiently.
- Waste generation varies throughout the day.
- Renewable energy availability changes with weather.

A conventional dashboard can show this information, but it does not necessarily **reason about the problem and take action** .

Challenge

Develop an **AI-powered autonomous sustainability system** capable of monitoring campus resources, identifying inefficiencies and recommending or executing corrective actions.

The system should ideally combine:

IoT/Sensor Data + AI + Prediction + Reasoning + Automation

Possible use cases:

- Electricity optimization
- Smart classroom management
- Water-leak detection
- Waste classification
- Solar-energy optimization
- HVAC optimization
- Occupancy prediction
- Carbon-footprint estimation
- Smart parking
- Campus environmental monitoring

Expected Outcome

The prototype should:

1. Collect or simulate real-time campus data.
2. Detect abnormal or inefficient resource usage.
3. Predict future consumption where applicable.
4. Identify possible causes.
5. Recommend an optimization strategy.
6. Demonstrate an automated action or intervention.

For example:

Sensor detects abnormal electricity consumption → AI identifies probable cause → predicts continued wastage → agent recommends action → authorized system executes/requests intervention.

Innovation Opportunities

- IoT
- Edge AI
- Time-series forecasting
- Computer vision
- Digital twins
- Reinforcement learning
- AI agents
- Optimization algorithms

- Predictive maintenance



Suggested Evaluation Framework

For a national-level event, I would use the same broad judging framework for all four challenges:

Criterion	Weight
Problem Understanding & Relevance	10%
AI/ML Innovation	20%
Technical Implementation	20%
Reasoning / Autonomous Capability	15%
Real-world Impact & Scalability	15%
Accuracy / Performance	10%
UX & Demonstration	5%
Responsible & Ethical AI	5%
Total	100%